Universal Compression Architecture

Benjamin Walker Mayes

Unified Dynamics Alliance

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Abstract

We define a *compression system* as a tuple K = (I, C, O, tau, rho, delta) and identify six structural features (F1-F6) that characterize how such systems compress unbounded input into transmissible, identity-preserving output. We show that every natural language, analyzed as a compression system, exhibits a proper subset of these six features, while mathematics is the unique system exhibiting all six. The six features map bijectively onto the six core transport concepts of Unified Dynamics (UD): spin, stationarity, degenerate repair, factored representation, convergent simplification, and intentional superposition. A compression universality corollary conjectures that the feature space is the universal solution space for compression under identity preservation. This result is a structural correspondence, not a claim that languages are physics.

Keywords: compression architecture, loop spine, natural language, eigenspace decomposition, transport features, information theory, structural isomorphism

I. INTRODUCTION

Every natural language solves the same fundamental problem: compress unbounded experience into transmissible structure that preserves identity under iteration. The strategies differ across languages and eras, but the underlying constraint is universal. In this paper we formalize this constraint as a six-feature architecture and show that it maps bijectively onto the transport concepts of Unified Dynamics.

The motivation is not to claim that languages are physics, but to identify the shared logical skeleton that both linguistic and physical systems must satisfy when compressing unbounded input into finite, recoverable structure. The UD loop spine [THM-LS1-LS12] provides the formal vocabulary; the natural languages provide the empirical test cases.

II. DEFINITIONS

Definition 1 (Compression System). A *compression system* is a tuple

$$K = (I, C, O, tau, rho, delta)$$

where I is an unbounded input space, C is a finite carrier space, O is an output space of recoverable meaning, tau : I --> C is a transport map (encoding), rho : C --> C is a closure map (identity preservation under iteration), and delta : C x C --> R^+ is a repair metric (cost of restoring corrupted carriers).

Definition 2 (Compression Features). A compression system K may exhibit any subset of the following six structural features:

(F1) Dual-mode carriage: A single carrier c supports two independent information channels.

(F2) Positional eigenvalue: The meaning of c depends on its position in a sequence, not on internal modification.

(F3) Degenerate free repair: Multiple distinct arrangements yield the same output, with delta(c_{sigma}, $c_{sigma'}$) = 0 for all permutations within the degenerate class.

(F4) Factored representation: Meaning decomposes as M = R x P where R (root/structure) and P (pattern/transport) are independent.

(F5) Convergent simplification: Under repeated transmission, the system trends toward a lower-dimensional carrier subspace.

(F6) Intentional superposition: The carrier representation deliberately preserves multiple outputs simultaneously; a selection operation projects onto a single reading.

Definition 3 (Loop-Spine Completeness). A compression system K is *loop-spine complete* if it exhibits all six features (F1)-(F6) simultaneously.

III. THEOREM

Theorem (Universal Compression Architecture, THM-570).

(a) Every natural language, analyzed as a compression system K, exhibits at least one and at most five of the features (F1)-(F6). No natural language exhibits all six.

(b) Mathematics, analyzed as a compression system, is the unique system achieving full reconstructibility, corresponding to the full loop spine L_R.

(c) The six features map bijectively onto the six core UD transport concepts.

(d) Specific languages exhibit the feature subsets classified in Table I.

TABLE I. Feature-to-UD-concept bijection.

Feature	UD Concept	Loop-Spine Element
(F1) Dual-mode	Spin operator S	Edge carries sigma, sigma-bar
(F2) Positional	T-eigenstate stationarity	V1: position determines role
(F3) Free repair	B = 0 in degenerate sectors	V1/3: free rearrangement
(F4) Factored	Layer A / Layer B separation	Structure vs. transport
(F5) Convergent	T^n --> P1	THM-LS10: convergence
(F6) Superposition	Degenerate eigenspace	Multiple states, one eigenvalue

TABLE II. Language classification by compression features.

Language	Features	UD Eigenspace Analog
Sumerian	F1	S (spin)
Classical Chinese	F2	V1 (stationary)
Sanskrit	F1, F3	V1/3 (confined)
Classical Arabic	F3, F4	Binary orbit / Layer A-B
Classical Greek	F1, F2, F4	(T, S, *) triple
Latin	F3, F5	V1/3 --> V1 decay
Japanese	F1, F2, F4	(2,1,1,2) chirality
English	F1, F2, F5	V1/2 (broken)
Hebrew	F3, F4, F6	Superposition
Mathematics	F1-F6 (at LR)	Full loop spine

IV. PROOF

A. Part (a): Feature verification

Sumerian (F1): Cuneiform signs function simultaneously as logograms (semantic) and phonograms (phonetic). The sign AN reads as "sky" or as the syllable /an/. One carrier, two independent channels.

Classical Chinese (F2): An isolating language with no inflectional morphology. Meaning is determined by sentential position and tonal contour, not by modification of the character.

Sanskrit (F1, F3): The eight-case system permits nearly free word order. "Rama sees Sita" and "Sita Rama sees" carry identical content because case endings disambiguate. Repair cost for permutation is zero (F3). Sandhi rules constitute dual-mode boundary interaction (F1).

Classical Arabic (F3, F4): The trilateral root system separates semantic content (root k-t-b = writing) from grammatical realization (pattern: kataba/kitab/maktaba). Factored representation (F4). Ten verb forms are degenerate over a fixed root (F3).

Classical Greek (F1, F2, F4): The verb system decomposes into aspect, mood, and voice -- three independent axes. Triple decomposition (F4) with positional sensitivity (F2) and dual active/middle readings (F1).

Latin (F3, F5): The ablative carries eight distinct functions in one formal slot (F3). Historical evolution Latin --> Romance shows systematic case loss toward analytic structure (F5).

Japanese (F1, F2, F4): Three writing systems create visual chirality (F1). Particle-based grammar uses positional marking (F2). Scripts factor semantic, grammatical, and foreign content independently (F4).

English (F1, F2, F5): Dual Germanic/Romance vocabulary provides two modes for the same meaning (F1). Rigid SVO order determines role positionally (F2). Loss of Old English inflection shows convergent simplification (F5).

Hebrew (F3, F4, F6): Shares the trilateral root system with Arabic (F3, F4). The Torah's unvoweled text preserves multiple readings simultaneously; vocalization projects onto a single meaning (F6).

No natural language in the survey exhibits all six features. Each instantiates a proper subset.

B. Part (b): Mathematics as full L_R

In mathematics, every well-formed statement can be traced to axioms via a finite proof chain: tau is explicit (formal definition), rho is perfect (proved theorems remain proved), and delta = 0 for valid proofs. All six features are present: notation is dual-mode (F1), position matters in formulas (F2), equivalent reformulations cost zero (F3), content separates from notation (F4), complex proofs reduce to axioms (F5), and variables represent universally quantified superpositions until instantiated (F6). This is the loop spine at L_R: full transport, full closure, full reconstruction.

C. Part (c): Bijection

The bijection is established by construction in Table I. Each feature corresponds to exactly one UD concept. Independence: no feature implies another (verified by the existence of languages exhibiting each feature without the others). The UD concepts are mutually independent (established by THM-LS1-LS12). QED

V. COROLLARIES

Corollary 1 (Participation Map). The language classification in Table II is an instance of the participation map (A7) from THM-569: multiple compression systems participate in the same structural architecture, each instantiating a different projection. No natural language is the architecture; each is a shadow of it.

Corollary 2 (Compression Universality). Any system that compresses unbounded input into transmissible, identity-preserving, reconstructible output -- whether linguistic, physical, biological, or computational -- must instantiate at least one of the features (F1)-(F6). The feature space is the universal solution space for the

compression-under-identity problem.

VI. DISCUSSION

The universality of the pattern is not that languages are physics. It is that languages and physics are both solutions to the same problem: how to compress unbounded experience into transmissible, preservable, reconstructible structure. The answer, in every case, involves some combination of transport (propagation), closure (return/identity), reconstruction (repair after perturbation), eigenspace decomposition (separation into modes), degeneracy (free internal rearrangement), and convergence (simplification over time).

The practical implication is that the UD loop spine may serve as a diagnostic framework for analyzing compression systems in any domain. In particular, the application to large language models (LLMs) reveals that transformers instantiate all six features at the closure level (L_C) but not at the reconstruction level (L_R), identifying the hallucination problem as a structural gap rather than a training deficiency.

VII. SCOPE AND NON-CLAIMS

This theorem does not claim that languages are physics, that linguistic features cause physical phenomena, that the mapping exceeds structural analogy, or that every language classification is definitive (borderline cases exist). It does claim that the compression problem has a constrained solution space, that the UD loop spine characterizes that space, that each major language independently discovered a different subset of the features, and that mathematics is the unique system achieving full L_R.

Proof-state. Feature definitions (F1)-(F6): exact (mathematical). Language verifications: interpretive (linguistic analysis). Bijection: structural (construction). Compression universality: conjectural (completeness assertion).

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